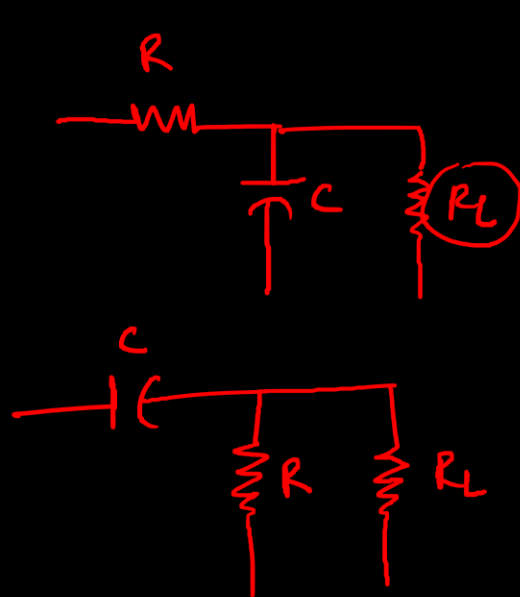
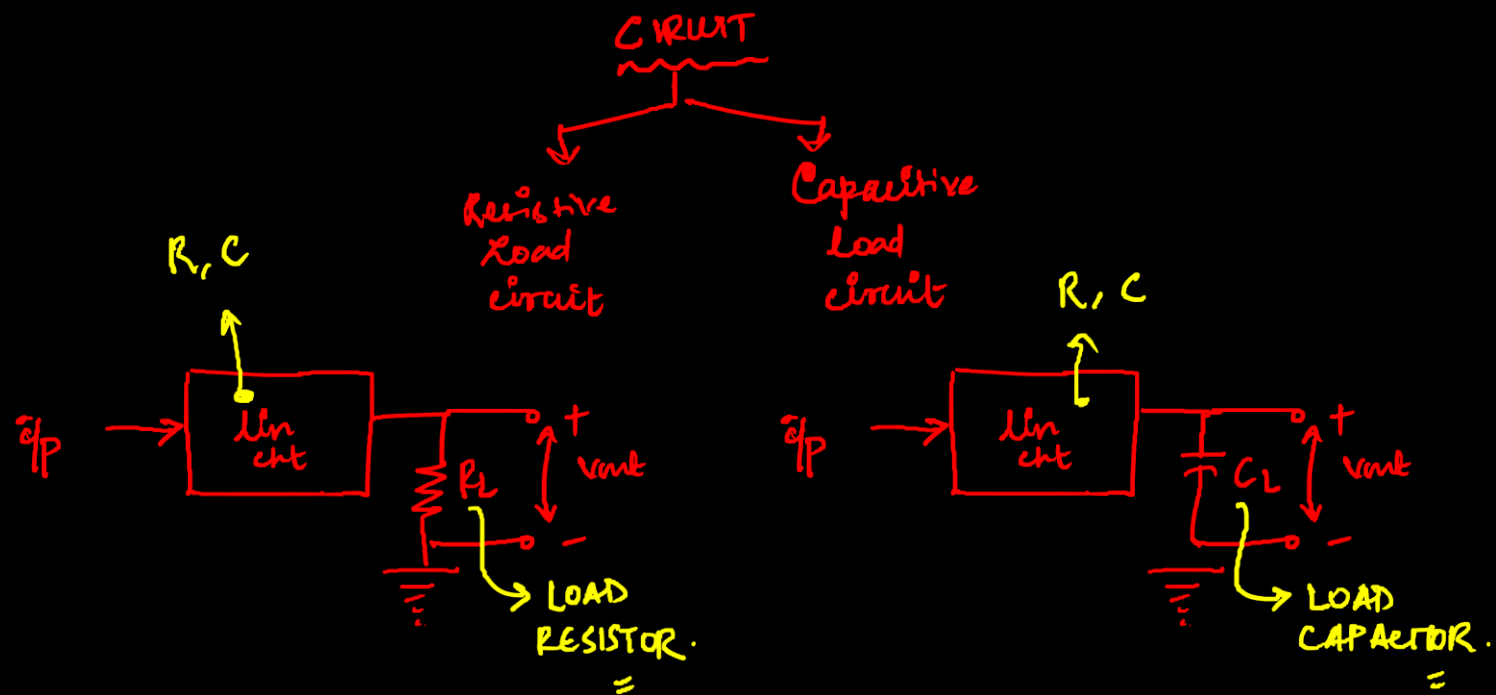


QUALITATIVE ANALYSIS OF RC CIRCUITS | 15-08-2025

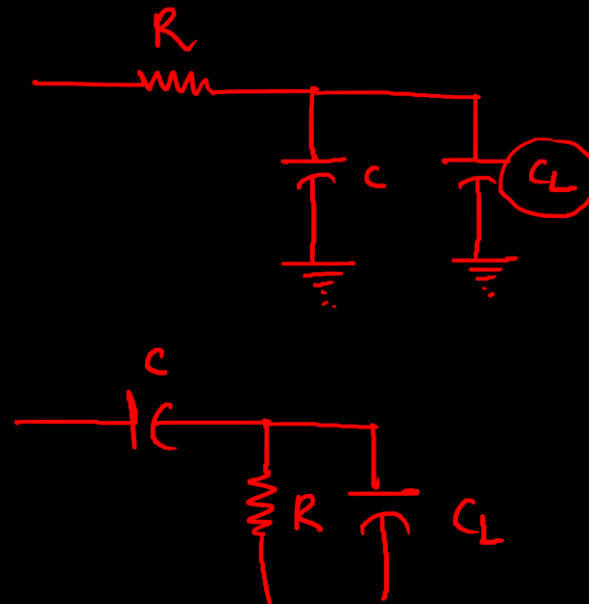
Er. Sailesh Mishra

B.Tech (ECE | Silicon University)

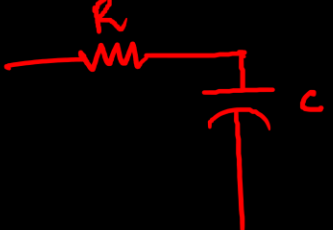
Specialization in Microelectronics & VLSI



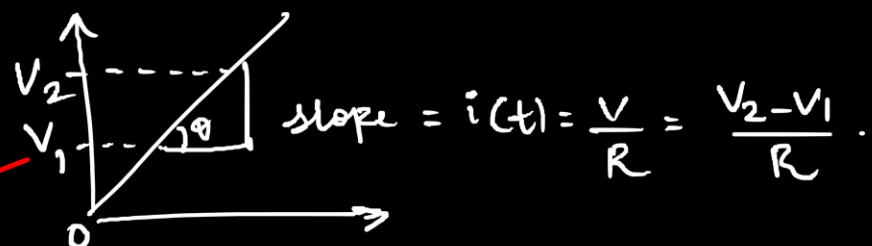
HIGH-PASS-FILTER.



LOW-PASS-FILTER.

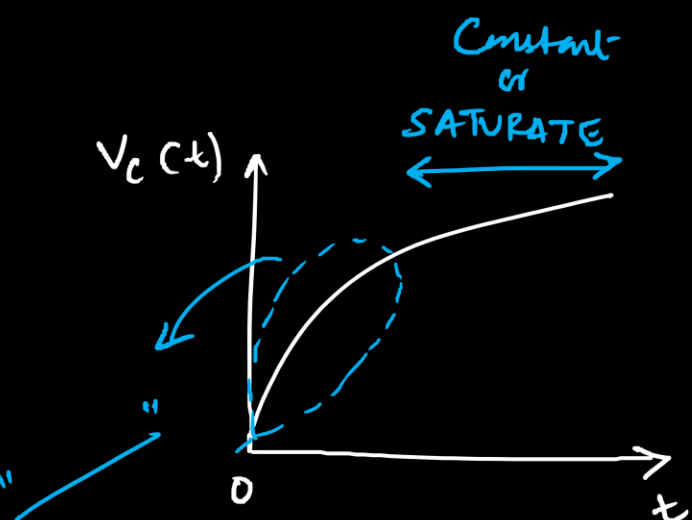


$\odot \quad i(t)_R = \frac{V}{R} = \frac{1.8}{1k\Omega} = 1.8 \text{ mA}$



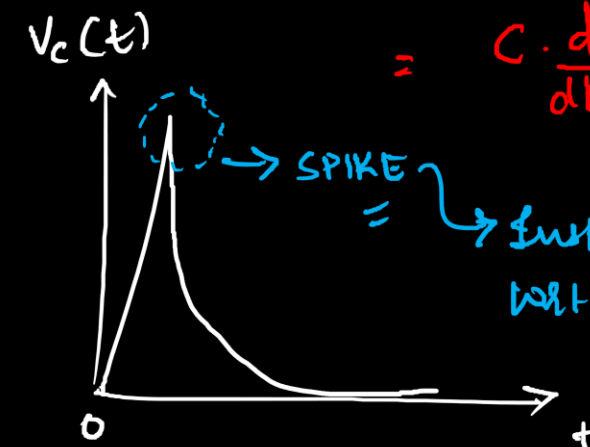
$\odot \quad i(t)_C = C \cdot \frac{dV}{dt} = C \cdot \left\{ \begin{array}{l} \text{Instantaneous change of} \\ \text{Input voltage w.r.t 't'} \end{array} \right\} = C \cdot \frac{d}{dt} \left(\frac{0}{1} \right) = C \times 0 = 0 \Rightarrow i(t) = 0 \text{ Means}$
 $R = \infty \rightarrow 0 \cdot C !!$

$= C \cdot \frac{d}{dt} \left(\frac{1}{0} \right) = C \times \infty = \infty \Rightarrow i(t) = \infty \text{ Means}$
 $R = 0\Omega \rightarrow \infty \cdot C !!$



"Linear graph = ohmic character = $V=IR$ "

R-C CIRCUIT
 [Low pass Filter]



SPIKE = Instantaneous voltage change w.r.t a very very small interval of time.

$\Delta t \rightarrow \leftarrow V \uparrow \uparrow$

C-R CIRCUIT
 [High pass Filter]

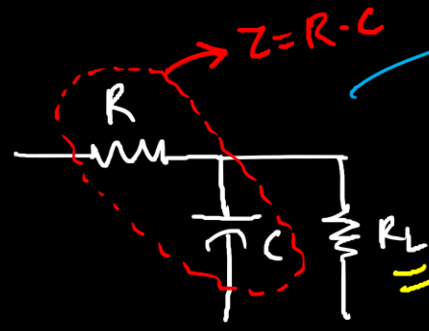
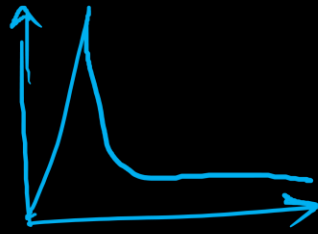
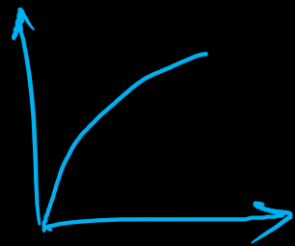
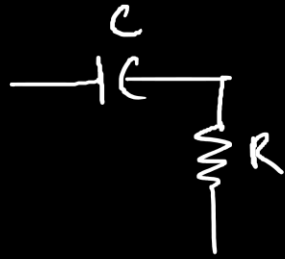
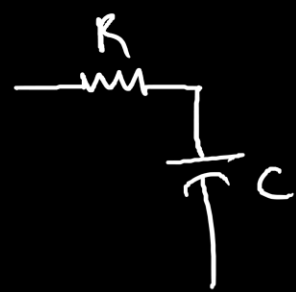
◆ TYPES OF FILTERS/RL circuits :-

(i) Low pass Filter $\rightarrow$ Series R-C ✓

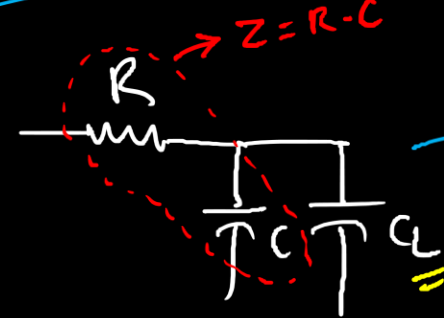
(ii) High pass Filter $\rightarrow$ Series C-R ✓

$\left\{ \begin{array}{l} \rightarrow \text{RESISTIVE LOAD } R-C \rightarrow \text{RESISTIVE LOAD LOW PASS FILTER} \\ \rightarrow \text{RESISTIVE LOAD } C-R \rightarrow \text{RESISTIVE LOAD HIGH PASS FILTER} \end{array} \right.$

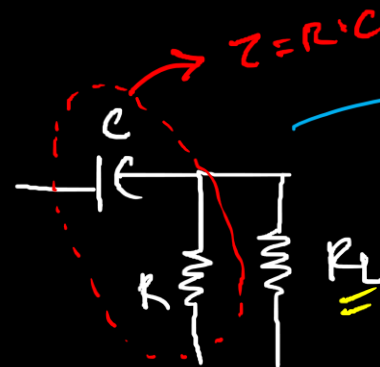
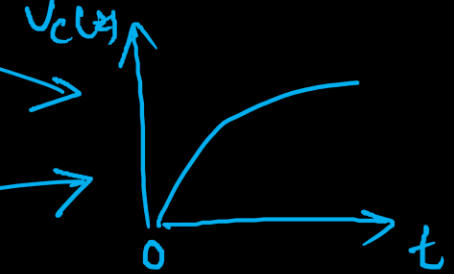
$\left\{ \begin{array}{l} \rightarrow \text{CAPACITIVE LOAD } R-C \rightarrow \text{CAPACITIVE LOAD LOW PASS FILTER} \\ \rightarrow \text{CAPACITIVE LOAD } C-R \rightarrow \text{CAPACITIVE LOAD HIGH PASS FILTER} \end{array} \right.$



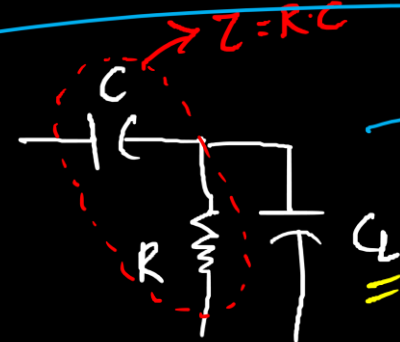
RESISTIVE LOAD



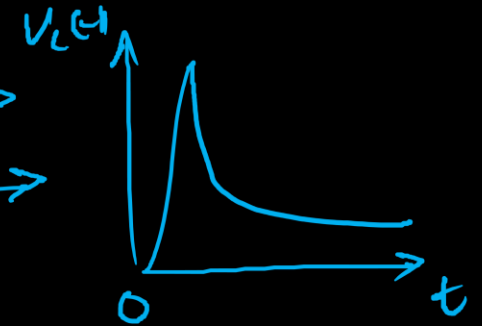
CAPACITIVE LOAD

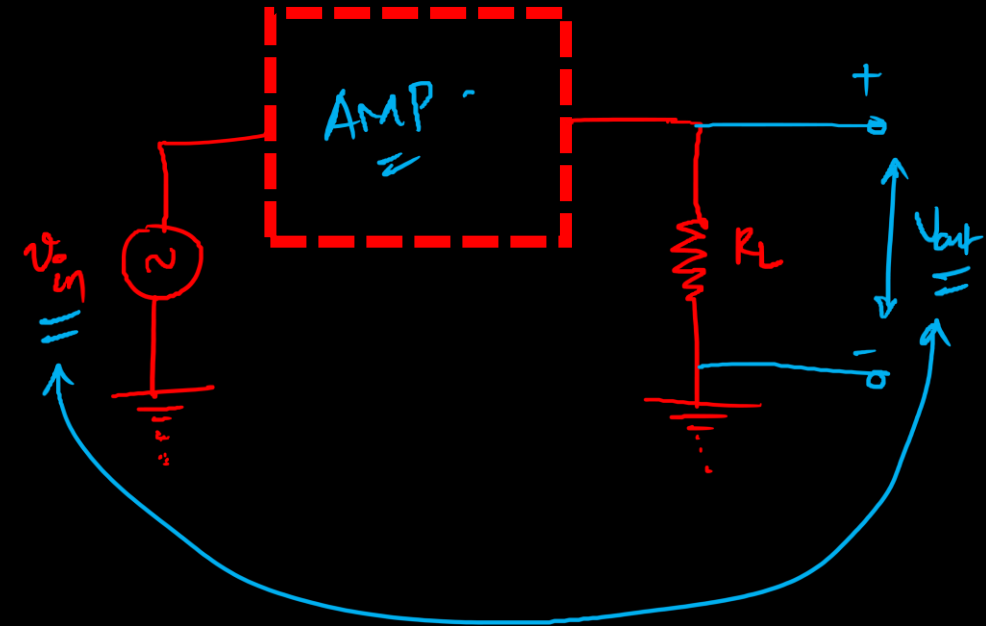
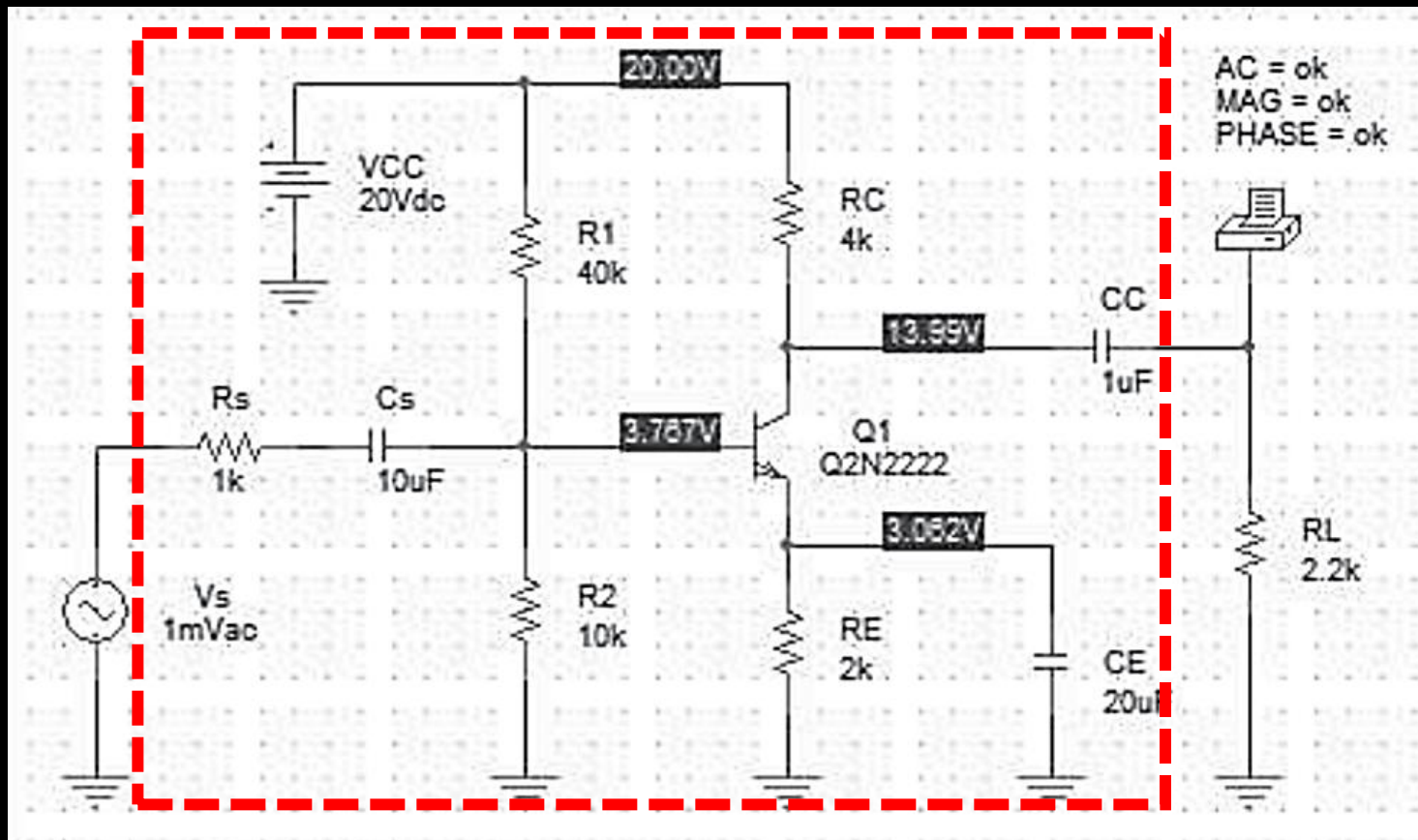


RESISTIVE LOAD



CAPACITIVE LOAD

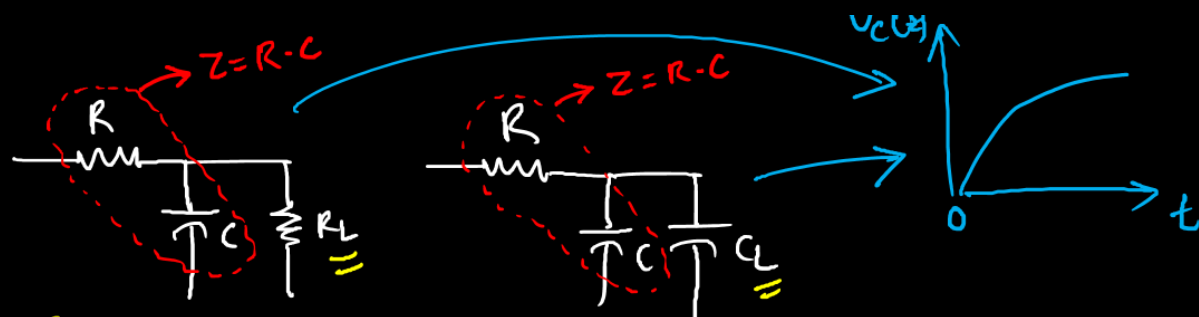




$$\rightarrow A_v = \frac{V_{out}}{V_{in}} \checkmark$$

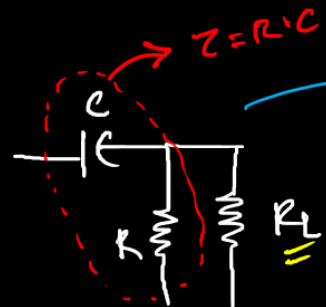
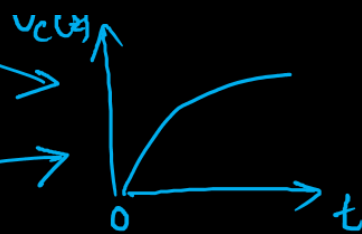
$$\rightarrow V_{out} = i_L \times (R_L)^{2.2K}$$

$$= i_C \times R_L$$

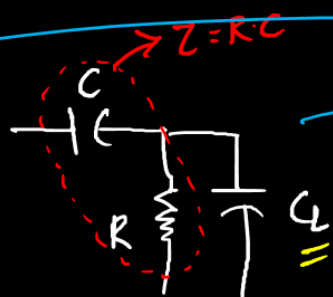


RESISTIVE LOAD

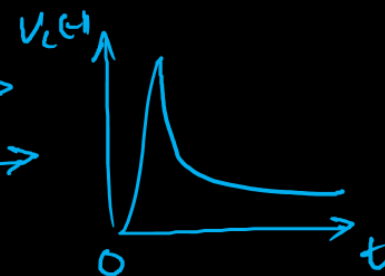
CAPACITIVE LOAD



RESISTIVE LOAD



CAPACITIVE LOAD



$R_L \rightarrow$ Power Consumption is Dominant
 $\rightarrow V_{in} \rightarrow i_L \downarrow =$!!

$\rightarrow C_L \rightarrow$ time delay $i_L \uparrow$

